Examiner only

## SECTION A

Answer **all** questions.

**1.** (a) Radon is a radioactive gas and emits alpha particles.

(i) Explain what is meant by the *activity* of a radioactive source **and** give its unit. [2]

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(ii) The half-life of radon is $3.3 \times 10^5$ s. Determine the time it takes for a sample of radon to decay to 20 % of its initial activity. [3]

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(iii) Discuss the extent to which alpha particles are a risk to the human body. [3]

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(b) Radium-226 decays into radon.

$$^{226}_{88}\text{Ra} \longrightarrow {}^{222}_{86}\text{Rn} + {}^{4}_{2}\alpha$$

Assuming that 98% of the energy released in this process is converted to the kinetic energy of the alpha particle, use the data below to determine the speed of the alpha particle. [5]

Atomic masses: $^{226}_{88}\text{Ra}$ = 226.025 410 u  $^{222}_{86}\text{Rn}$ = 222.017 578 u

$^{4}_{2}\text{He}$ = 4.002 603 u  $m_{\text{electron}}$ = 0.000 549 u

1 u = 931 MeV

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